

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		Momentum before = momentum after (1) provided no external forces act (1)	2			2		
		(ii)		No because of gravity Don't accept air resistance			1	1		
	(b)			Some is converted to heat or sound during collision / it's not an elastic collision / it's an inelastic collision / the block becomes deformed (1) and then into PE as the block swings up (1)			2	2		
	(c)	(i)		PE = KE or $\frac{1}{2}mv^2 = mgh$ or implied (1) $v^2 = 2gh$ (1) $= 2 \times 10 \times 0.119 = 2.38 \text{ [m}^2\text{/s}^2\text{]} (1)$ Velocity = 1.54 [m/s] (1) Alternative: $v^2 = u^2 + 2ax$ or implied (1) $0 = u^2 - 2 \times 10 \times 0.119 (1)$ $u^2 = 2.38 \text{ [m}^2\text{/s}^2\text{]} (1)$ Velocity = 1.54 [m/s] (1) Award 3 marks for answer of 15.4 [m/s]		4		4	3	
		(ii)		Attempt at: momentum of bullet = momentum of block and bullet (1) Conversion of 16 g into 0.016 kg (1) $0.016u = 2.516 \times 1.54$ (ecf) (1) $u = 242.2 \text{ [m/s]}$ Accept 242.6 [m/s] (1) Award 3 marks for answer of 0.242 or 1.782 [m/s]	1	1 1 1		4	3	
				Question 7 total	3	7	3	13	6	0